

Section „2“

Geometry

① Distance between two Given points

Let P and Q be two points with coordinates

$$P(x_1, y_1) \quad Q(x_2, y_2)$$

$$\Rightarrow \overline{PQ} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Note 1.1.1: The distance of P from the origin O is $OP = \sqrt{x_1^2 + y_1^2}$

Example 1.1.1

If P is the point $(4, 7)$ and Q is $(2, 3)$, then

$$PQ = \sqrt{(4-2)^2 + (7-3)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5} \text{ units}$$

Example 1.1.2

The distance between the points $P(2, -5)$ and $Q(-4, 7)$ is

$$PQ = \sqrt{(2+4)^2 + (-5-7)^2} = \sqrt{36+144} = \sqrt{180} = 6\sqrt{5} \text{ units}$$

Example =

Show that the points $(6, 6)$, $(2, 3)$ and $(4, 7)$ are the vertices of a right angled triangle.

Solution

Let A, B, C be the points $(6, 6)$, $(2, 3)$ and $(4, 7)$, respectively.

$$AB^2 = (6-2)^2 + (6-3)^2 = 16 + 9 = 25$$

$$BC^2 = (2-4)^2 + (3-7)^2 = 4 + 16 = 20$$

$$AC^2 = (6-4)^2 + (6-7)^2 = 4 + 1 = 5$$

$$AC^2 + BC^2 = 20 + 5 = 25 = AB^2$$

Hence, the points are the vertices of a right angled triangle.

Example :

Show that the set of points $(-2, -1)$, $(1, 0)$, $(4, 3)$ and $(1, 2)$ are the vertices of a parallelogram.

Solution

Let A, B, C, D be the points $(-2, -1)$, $(1, 0)$, $(4, 3)$ and $(1, 2)$, respectively. A quadrilateral is a parallelogram if the opposite sides are equal.

$$AB^2 = (-2 - 1)^2 + (-1 - 0)^2 = 9 + 1 = 10$$

$$BC^2 = (1 - 4)^2 + (0 - 3)^2 = 9 + 9 = 18$$

$$CD^2 = (4 - 1)^2 + (3 - 2)^2 = 9 + 1 = 10$$

$$AD^2 = (-2 - 1)^2 + (-1 - 2)^2 = 9 + 9 = 18$$

$$AB = CD = \sqrt{10} \quad BC = AD = \sqrt{18}$$

Since the opposite sides of the quadrilateral $ABCD$ are equal, the four points form a parallelogram.

Example :

Prove that the points $(3, 2)$, $(5, 4)$, $(3, 6)$ and $(1, 4)$ taken in order form a square.

Solution

A quadrilateral in which all sides are equal and diagonals are equal is a square. Let A, B, C, D be the points $(3, 2)$, $(5, 4)$, $(3, 6)$, $(1, 4)$, respectively.

$$AB^2 = (3 - 5)^2 + (2 - 4)^2 = 4 + 4 = 8 \quad CD^2 = (3 - 1)^2 + (6 - 4)^2 = 4 + 4 = 8$$

$$BC^2 = (5 - 3)^2 + (4 - 6)^2 = 4 + 4 = 8; \quad AD^2 = (3 - 1)^2 + (2 - 4)^2 = 4 + 4 = 8$$

$$AC^2 = (3 - 3)^2 + (2 - 6)^2 = 0 + 16 = 16 \quad BD^2 = (5 - 1)^2 + (4 - 4)^2 = 16 + 0 = 16$$

$$AB = BC = CD = AD = \sqrt{8} \quad AC = BD = \sqrt{16} = 4$$

Thus, all sides are equal and also the diagonals are equal. Hence, the four points form a square.

Example:-

Show that the points (3, 7), (6, 5) and (15, -1) lie on a straight line.

Solution

Let $A(3, 7)$, $B(6, 5)$ and $C(15, -1)$ be the three points. Then

$$AB^2 = (3 - 6)^2 + (7 - 5)^2 = 9 + 4 = 13$$

$$BC^2 = (6 - 15)^2 + (5 + 1)^2 = 81 + 36 = 117$$

$$AC^2 = (3 - 15)^2 + (7 + 1)^2 = 144 + 64 = 208$$

$$\therefore AB = \sqrt{13}$$

$$BC = \sqrt{117} = \sqrt{9 \times 13} = 3\sqrt{13}$$

$$AC = \sqrt{208} = \sqrt{16 \times 13} = 4\sqrt{13}$$

$$AB + BC = AC = 4\sqrt{13}$$

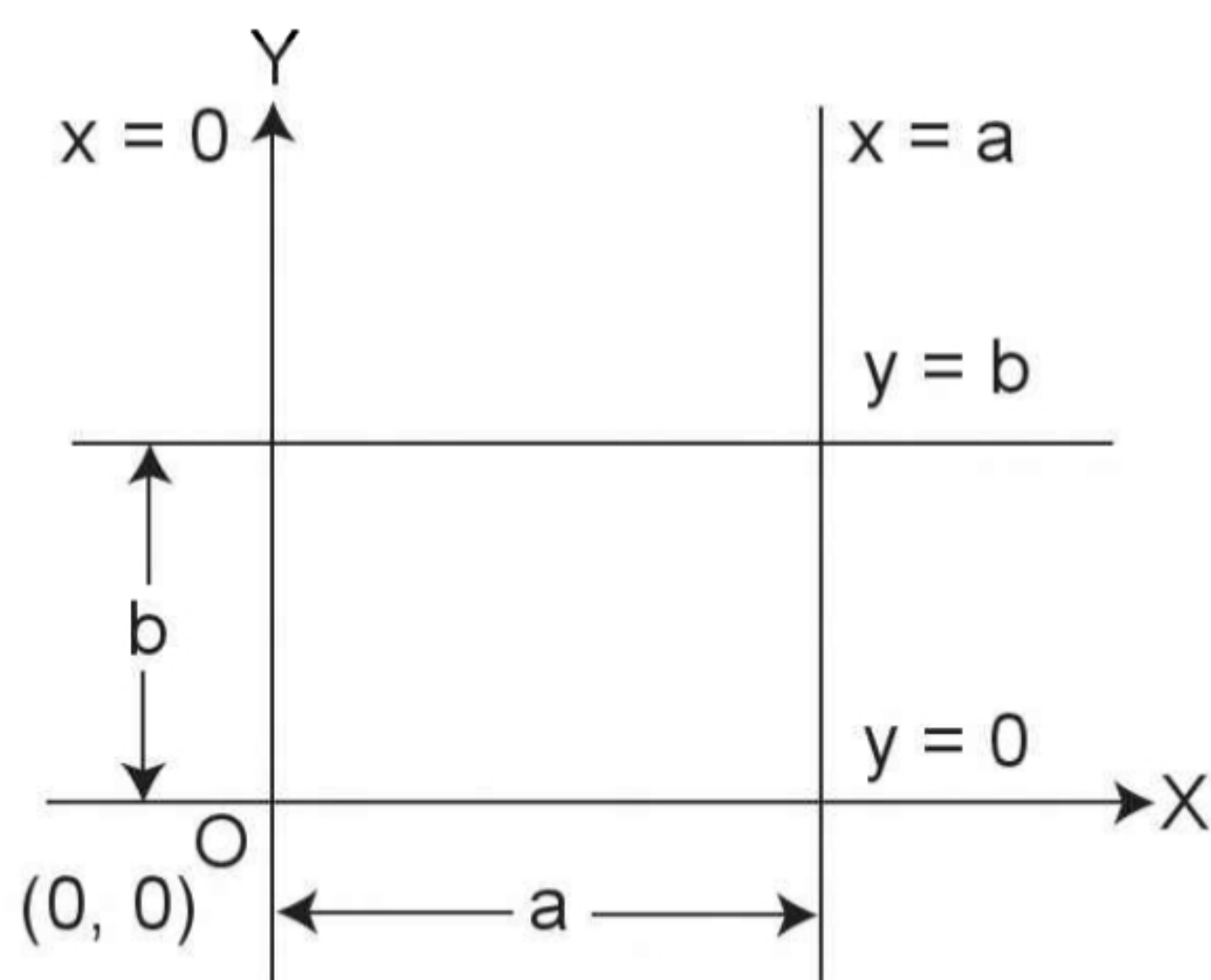
Hence, the three given points lie on a straight line.

Straight line Equations

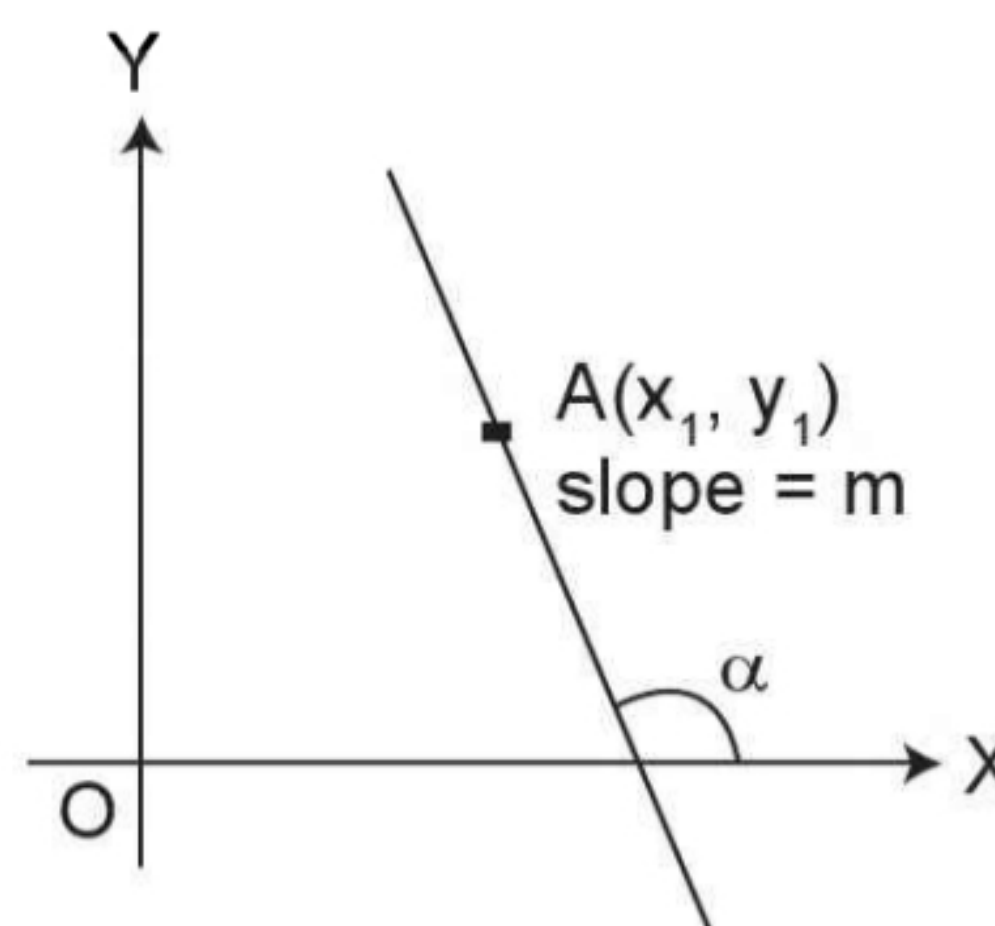
An equation (First degree in x and y) of the form $ax + by + c = 0$ always represents a straight line

Various forms of equation of straight line

1. The equation of x -axis is $y = 0$.
2. The equation of y -axis is $x = 0$.
3. The equation of a straight line parallel to x -axis at a distance b from origin is $y = b$.
4. The equation of a straight line parallel to y -axis at a distance a from origin is $x = a$.



2. **Point Slope Form** The equation of straight line, which has slope m and passes through a fixed point $A(x_1, y_1)$, is:
 $y - y_1 = m(x - x_1)$.



$$m = \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

- 2.1 The equation of a line through the point (x_1, y_1) and parallel to x -axis is $y = y_1$.
- 2.2 The equation of a line through the point (x_1, y_1) and parallel to y -axis is $x = x_1$.
3. Any non-vertical line through origin having slope m is $y = mx$

Normal form

Divide the equation (i) by $\sqrt{a^2 + b^2}$, we get,

$$\frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \pm \left(\frac{-c}{\sqrt{a^2 + b^2}} \right)$$

out of $\pm$ sign we consider that sign which will make right hand side + ve (positive).

straight line passing through two points

$(x_1, y_1), (x_2, y_2)$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example = Find the equation of the straight line passing through $(3, -2), (1, 7)$

Sol

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\Rightarrow \frac{y + 2}{x - 3} = \frac{7 + 2}{1 - 3}$$

$$-2(y + 2) = 9(x - 3)$$

$$\Rightarrow -2y - 4 = 9x - 27$$

$$\Rightarrow 9x + 2y - 23 = 0$$

let the two points $(a, 0), (0, b)$

$$\frac{y - 0}{x - a} = \frac{b - 0}{0 - a}$$

$$\Rightarrow \frac{y}{x - a} = \frac{b}{-a} \Rightarrow -ay = bx - ab$$

divide by ab , we get $\Rightarrow bx + ay = ab$

$$\frac{x}{a} + \frac{y}{b} = 1 \rightarrow \underline{\underline{\text{The intercept form}}}$$

Example = Find the equation of the straight line passing through $(4, 0)$, $(0, 8)$

sol

we can use $\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$ or we can substitute directly in the form $\frac{x}{a} + \frac{y}{b} = 1$

$$\Rightarrow \frac{x}{4} + \frac{y}{8} = 1$$

$$\Rightarrow \underline{\underline{8x + 4y - 32 = 0}} \quad \text{or} \quad \underline{\underline{2x + y - 8 = 0}}$$

Two lines are perpendicular when $m_1 m_2 = -1$
and are parallel when $m_1 = m_2$

* The intercept slope form =

$$y = mx + c$$

$m \equiv \text{slope}$

$c \equiv (0, c)$
↓
intercept (y-axis)

to prove =

we know = $y - y_1 = m(x - x_1)$

$$\Rightarrow y - c = m(x - 0)$$

$$\Rightarrow y - c = mx$$

$$\Rightarrow \boxed{y = mx + c}$$

Example: $m = 2$, intercept on y-axis = 3
Sol

we know $y = mx + c$

$$\Rightarrow y = 2x + 3$$

$$\Rightarrow 2x - y + 3 = 0$$

Examples =

① The line $6x + 5y + 7 = 0$ passing through $(k, -2)$, find k

② Find the equation of the line passing through the intersection point of $2x - 3y = 8$
 $5x - y = 7$

and its slope is 2.

③ prove that $3x + 4y - 12 = 0$
 $6x + 8y + 31 = 0$
are parallel.

④ find straight line equation, such that
it passes through (5, 7) and parallel to

$$4x + 3y - 6 = 0$$

⑤ Find the equation of the straight line
passes through intersection point of $2x + 3y = 5$
 $x + y = 2$
and perpendicular to $x - 2y + 9 = 0$